

Coherence-Selection Events and Causal Sets in Modal Triplet Theory:

Non-Zeno Conditions, Reset Laws, and Covariant Statistics

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Version 2

Abstract

Coherence-selection dynamics can generate an effective causal set only when the event law prevents accumulation and is covariantly defined. This paper states two sufficient routes. In a deterministic hybrid model, a reset margin together with a uniform bound on trigger growth gives a positive dwell time and hence finitely many events on every bounded trajectory interval. In a stochastic model, finite compensator measure on every compact spacetime region implies almost-sure local finiteness. Once local finiteness is established, events inherit a partial order from a globally hyperbolic spacetime. Positive stability margins alone do not provide either result, and noninvertible projection alone does not define event times, reset states, outcome probabilities, or irreversibility. Lorentz-invariant statistics require a concrete covariant event law, such as a scalar intensity with respect to spacetime volume; they do not follow from coherence language. The current MTT corpus does not yet select the required trigger, reset, or hazard. The construction is therefore a rigorous conditional event encoding, not a derivation of fundamental spacetime discreteness or measurement outcomes.

Version 2 Revision Note

Supersedes	Version 1.0, <i>Causal Sets as Event-Selection Shadows of Coherence Break-down</i> .
Reason	The original paper inferred local finiteness from positive stability margins, left the transition/reset law undefined, and claimed approximately Lorentz-invariant statistics without a stochastic source.
Resolution	Deterministic dwell-time and stochastic compensator criteria are proved, the event/reset data are typed explicitly, and covariance is made a property of the selected law.
Retained result	Dynamically generated, locally finite events inherit the causal order of the effective spacetime and can be represented as a causal set.
Remaining boundary	MTT must select the trigger or hazard, reset law, spacetime density, and any outcome instrument from the same physical branch.

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1 Events need a law

An “event” is not created by naming a loss of coherence. A dynamical event model must specify:

- a state space and continuous evolution between events;
- an event trigger or stochastic hazard;
- the spacetime location assigned to an event;
- a reset or transition rule after the event;
- whether the event is an observed record, an objective state change, or only a coarse-graining marker.

These distinctions matter in MTT. A noninvertible lower description can discard information without selecting a unique physical outcome. Likewise, a stable fixed point can exist without any finite sequence of transition times.

2 Deterministic hybrid route

Let $z(t)$ be a continuous state between event times

$$0 < t_1 < t_2 < \cdots .$$

Let $h(z)$ be a scalar trigger. An event occurs when h reaches zero from below, after which a reset map R replaces $z(t_n^-)$ by $z(t_n^+) = R(z(t_n^-))$.

Theorem 2.1 (Uniform dwell-time criterion). *Assume that after every event*

$$h(z(t_n^+)) \leq -\delta$$

for a fixed $\delta > 0$, and that between events

$$\left| \frac{d}{dt} h(z(t)) \right| \leq M$$

for a fixed finite $M > 0$. Then

$$t_{n+1} - t_n \geq \frac{\delta}{M}.$$

Consequently a trajectory has at most

$$1 + \frac{MT}{\delta}$$

events in any interval of duration T .

Proof. After the reset, the trigger must increase by at least δ before it can reach zero. The derivative bound implies that this takes at least δ/M . Summing the minimum separations gives the counting bound. $\square$

This is the missing non-Zeno theorem. A positive stability margin for a fixed point does not imply either a reset margin or a bound on trigger growth.

Example 2.2 (Stability without a dwell-time theorem). The continuous system $\dot{x} = -x$ has a positive linear stability rate at the origin. If one separately declares events at $t_n = 1 - 2^{-n}$, the events accumulate at $t = 1$. The flow's stability has not prevented the Zeno schedule because the event law was independent of the flow.

3 Stochastic hazard route

Let $N(K)$ count events in a measurable spacetime region K . A point process may be specified through a compensator or intensity measure Λ .

Theorem 3.1 (Finite-compensator local finiteness). *Suppose an event point process satisfies*

$$\mathbb{E}[N(K)] = \Lambda(K) < \infty$$

for every compact $K \subset Y_4$. Then $N(K) < \infty$ almost surely for every member of a countable compact exhaustion. In particular, the event set is locally finite almost surely.

Proof. A nonnegative extended-integer random variable that equals infinity with positive probability has infinite expectation. Therefore finite expectation implies $N(K) < \infty$ almost surely. Apply this to a countable compact exhaustion. $\square$

A sufficient form is

$$\Lambda(K) = \int_K \lambda(x, z) d\text{vol}_g(x),$$

with a nonnegative scalar predictable intensity λ that is locally integrable. A uniform compact-region bound on λ gives finite $\Lambda(K)$.

This theorem does not derive Poisson statistics. Correlated point processes, self-exciting processes, and state-dependent hazards can all satisfy local finiteness. Their physical predictions differ.

4 From events to a causal set

Let $C \subset Y_4$ be the generated event set. On a globally hyperbolic spacetime define

$$x \preceq y \iff y \in J^+(x).$$

The kinematic companion paper proves that any locally finite C with this inherited order is a causal set. Thus either [Theorem 2.1](#), with the required spatial population bound, or [Theorem 3.1](#) can provide the local-finiteness input.

For a deterministic family of trajectories, a compact spacetime region must intersect only finitely many relevant trajectories, or a separate uniform spatial density bound is needed. Dwell time along each of infinitely many worldlines would not by itself control the total number of events in the region.

5 Reset, irreversibility, and records

A reset map can be invertible or noninvertible. If it is noninvertible, the effective hybrid dynamics loses information. That is a model of irreversibility at the chosen descriptive level, not automatically a fundamental arrow of time.

For quantum states, an outcome event requires an instrument $\{\mathcal{I}_a\}$, not only an effect or decoherence channel. The probability and conditional state are

$$p_a = \text{tr}[\mathcal{I}_a(\rho)], \quad \rho_a = \frac{\mathcal{I}_a(\rho)}{p_a}$$

when $p_a > 0$. A nonselective channel $\sum_a \mathcal{I}_a$ does not say which event occurred. The current MTT Born-source result is closed only on a restricted selected recorder and remains open for arbitrary apparatus contexts and objective ontic selection.

Measurement has no privileged metaphysical role here. It is one physical process capable of producing records. The same event mathematics can describe switching, threshold crossings, detector clicks, or other transitions.

6 Covariance and Lorentz statistics

Lorentz-invariant or generally covariant statistics are properties of the event law. In Minkowski spacetime, a constant scalar Poisson intensity

$$\Lambda(B) = \rho \int_B d^4x$$

is invariant in law under proper orthochronous Poincare transformations. In curved spacetime, a scalar $\lambda(x, z)$ integrated against $d \text{vol}_g$ is covariantly defined if the state variables and evolution are also geometrically specified.

By contrast, a reset at equal coordinate-time intervals or a hazard tied to a preferred foliation generally breaks Lorentz symmetry. An approximately isotropic numerical event cloud is not a proof of invariant law.

7 Relation to MTT data

The local $1 + 2 + 3$ filtration, shared phase line, finite q79 Hessian, and fixed-point stability results do not currently emit:

- a four-dimensional scalar trigger h ;
- a reset map R ;
- a spacetime hazard λ ;
- a covariant event-location rule;

- an event density in physical units.

These objects could be derived in a future same-source event model. For example, a response functional might define a scalar threshold, while a selected detector or basin map defines the reset. But the cross-sector intertwiner and unit map must be exhibited. An internal spectral gap alone does not supply a four-volume intensity.

8 Status ledger

Statement	Status	Boundary
Reset margin plus derivative bound gives dwell time	Exact	Theorem 2.1.
Finite compensator gives local finiteness	Exact	Theorem 3.1.
Local events inherit causal-set order	Exact, imported	Kinematic companion theorem.
Positive stability margin excludes Zeno events	False in general	Event/reset law is additional.
MTT selects trigger/reset law	Open	No same-source event generator.
MTT predicts Lorentz-invariant event statistics	Open	Requires a covariant stochastic or hybrid law.
Event channel selects objective outcome	Open	Instrument, probability, and ontology are separate.

9 Discussion

The corrected event route is sharper than the original proposal. It gives two concrete mathematical exits from the local-finiteness blocker and makes clear what a simulation would need to test. A hybrid simulation should log the reset margin and maximum trigger derivative. A stochastic simulation should estimate or certify compact-region compensators and test covariance of the full law, not only the appearance of one sample.

The shortest path to an MTT-specific result is to construct one physical event source on the already selected free q79 QFT net. A detector instrument, a covariant hazard derived from its response kernel, and a finite-compensator proof would produce a genuine event causal set without assuming Poisson sprinkling.

10 Conclusion

Coherence-selection events can form a causal set, but only after non-accumulation and causal-order conditions are proved. Uniform reset margins and trigger bounds provide one route; finite stochastic compensators provide another.

MTT has not yet selected either event law. The present result is therefore a conditional mathematical framework. It replaces an unsupported local-finiteness claim with exact, testable criteria and preserves the distinction between effective irreversibility, observed records, and objective outcomes.

References

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